

NeurIPS 2022 | Cutting Label Cost for Table NER with Active Learning

Siemens and LMU Munich test active learning on a real industrial NER task, and find that built-in batch diversity helps while forced table diversity backfires

Industry runs on spreadsheets. The equipment, sensors, and vessels a plant operator tracks usually live in loosely structured tables, and pulling that information out automatically means fine-tuning a tabular language model on labeled examples. The catch is cost. These tables are written in highly technical, domain-specific language that only a handful of experts can read, let alone annotate, so assembling enough training labels quickly becomes prohibitively expensive.

Active learning is built for exactly this situation: plenty of unlabeled data, scarce and expensive labels, and a desire to extract as much model performance as possible from every annotation. Yet despite the obvious fit, no prior work had paired active learning with tabular language models. This workshop paper, by Martin Ringsquandl and Aneta Koleva of Siemens (Koleva is also at LMU Munich), presented at the Table Representation Learning Workshop at NeurIPS 2022, sets out to fill that gap.

Their task is sub-cell named entity recognition (NER) on tables: a single table holds many cells, and each cell may carry several token-level labels at once. That makes it an unusual, nested multi-instance active-learning problem, and it raises a question that has no obvious answer: at which level should you acquire labels, the token, the cell, or the whole table?

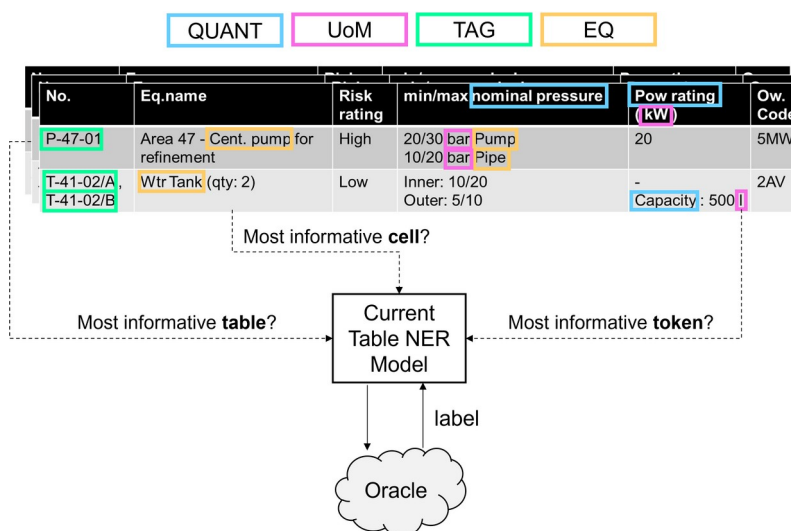


Figure 1. The industrial sub-cell NER problem and its acquisition: four entity types, QUANT, UoM, TAG, and EQ, are interleaved across a table's cells, and the model queries an expert oracle, with the open question of whether to pick the most informative token, cell, or whole table.

Paper: <https://arxiv.org/abs/2211.04128>

Where the labeling cost comes from

Fine-tuning a tabular language model for an industrial task needs many labels, and only experts who can read these specialized tables can provide them, which is what keeps acquisition costs high. The structure of the task makes it worse. Sub-cell NER is a nested multi-instance problem: a table contains many cells, and each cell carries multiple token labels, so the granularity at which you acquire annotations is itself an open question.

A second constraint comes from the model. A large transformer cannot be updated one example at a time the way classic active learning assumes; it needs batches, trained over several epochs per round, so the acquisition function has to select a whole batch at once. Pure uncertainty sampling tends to grab a cluster of highly correlated, redundant cells, often from the same table or even the same row, which wastes the annotation budget on near-duplicates.

The method: one tabular language model, four acquisition functions

The model is a table-biased transformer encoder. It replaces vanilla attention with a row-column visibility matrix and adds within-cell positional encoding, so each token gets a representation aware of the entire table. A classification decoder then tags every token using IO labels over four entity types: TAG (equipment identifiers), EQ (equipment names), QUANT (physical quantities), and UoM (units of measurement).

The authors make one key design argument: because the encoder already makes each cell aware of its full-table context, acquiring at the cell level is enough, and there is no need for a separate hierarchical table-then-cell selection. On top of that judgment they run standard pool-based active learning: each round selects a batch of the most informative cells, sends them to an expert oracle for labeling, and retrain on the enlarged set.

What actually gets compared are four cell-level acquisition functions. MNLP scores cells by their normalized log-probability uncertainty, a pure uncertainty strategy. MNLP+ adds a forced round-robin constraint that maximizes table diversity. BADGE runs k-MEANS++ over per-cell gradient embeddings to select a batch that is both uncertain and diverse. Rand samples cells uniformly at random as a baseline.

The data: a real industrial dataset

The evaluation uses a genuine industrial dataset rather than an academic benchmark. Spreadsheets from several plants were downsampled to at most five rows per table, then annotated cell by cell by experts using prodi.gy, a span-based annotation tool. A random split yields 55 training tables and 24 test tables. The hard part is how sparse and imbalanced the labels are.

Table 1. Training-set statistics.

Statistic	Value
#tables	55
#cells	4774
#labels	1112
labels per cell	0.23

The 55 training tables hold 4774 cells but only 1112 NER span labels, about 0.23 labels per cell. Put differently, roughly 77% of cells contain only O-tagged tokens, no entity at all, an extreme class imbalance the authors note is typical of real industrial data. Experiments start from a stratified seed of 100 labeled cells, add 50 cells per iteration, and average every result over 5 random seeds. The backbone is bert-base-uncased, fine-tuned for up to 30 epochs per round.

The result: BADGE stands out, forced diversity finishes last

Performance is measured by micro-averaged F1 on the held-out test set as active learning proceeds. The figure below plots the four acquisition functions together with the ceiling from training on the full dataset (full), against the cumulative number of labels. Two things are worth watching: which curve reaches the orange full line first, and how the methods rank against each other.

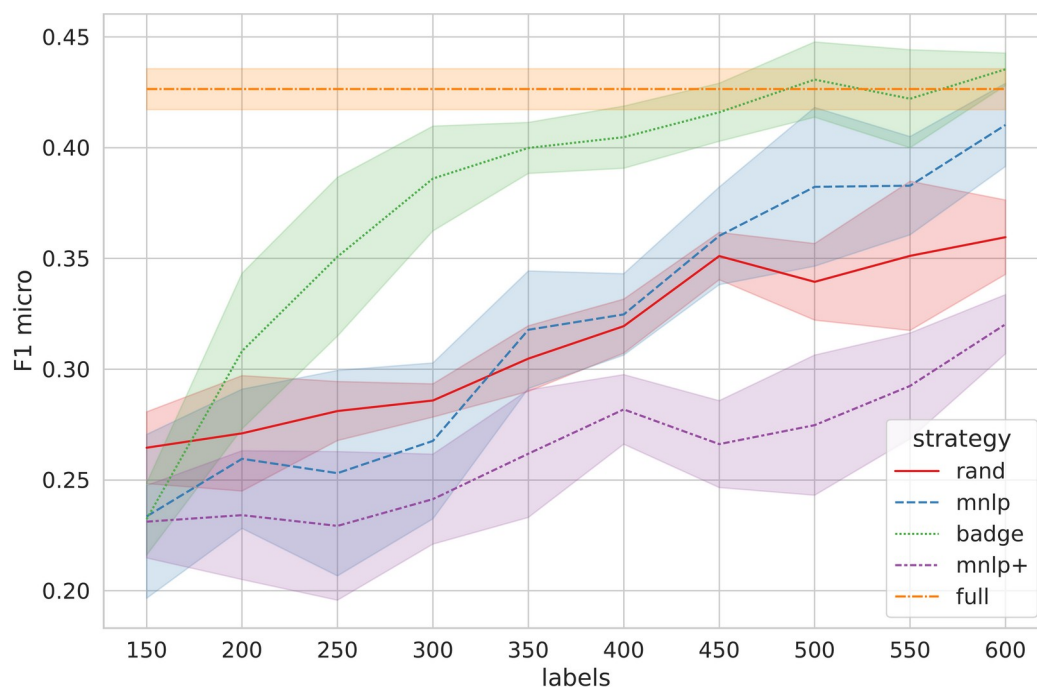


Figure 2. Micro F1 versus number of labels per acquisition function (averaged over 5 seeds): green badge reaches the orange full ceiling first and exceeds it, blue mnlp only edges past red rand after about 500 labels, and purple mnlp+ (forced maximum table diversity) trails throughout, even below random.

The picture is sharp. BADGE, the batch-diverse gradient method, is the only acquisition function that clearly beats random and even surpasses the full-training F1 ceiling, reaching that ceiling with far fewer labels. Pure uncertainty sampling with MNLP is disappointing: it only pulls slightly ahead of Rand after around 500 labels, and with noticeably higher variance, which suggests it is more sensitive to the initial seed in this highly imbalanced regime. Most striking is MNLP+, the variant that forces maximum table diversity: it is the worst of the four, falling below even the random baseline, which indicates that pushing diversity to the extreme actively hurts.

Why? Diversity is not a free good

The four acquisition functions effectively ablate the role of diversity. To see the mechanism, the authors track how many distinct tables each method samples from at every iteration: MNLP fixates on a few tables, MNLP+ spreads across as many as the budget allows, and BADGE sits in between. Lined up against the F1 results, the reading is clear. What wins is not maximum diversity but a careful trade-off between diversity and per-cell uncertainty: built-in batch diversity (BADGE) helps, while forced maximum table diversity (MNLP+) hurts.

Takeaway and boundaries

The lasting message is that active learning really can cut the expert-labeling cost of tabular language models, but only if diversity is handled with care. Cell-level acquisition with built-in batch diversity, as BADGE does it, reaches full-training performance with a fraction of the labels, whereas bluntly forcing maximum table diversity backfires and does worse than random. The practical recipe is to balance diversity against uncertainty rather than maximize either one alone.

As the first study of this problem, the paper is honest about what it leaves open. Two frontiers stand out: the computational cost of table-diverse acquisition, and the extra cognitive load it places on human annotators, who must reinterpret the context of each new table they are handed. Those are the questions that come next when moving active learning from a study into a real industrial annotation pipeline.